

ARKHANGELSK TALAGI, Russian Federation

WMO: 225500

Lat: 64.5042N

Lon: 40.7278E

Elev: 8

StdP: 101.23

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.2	-28.7	-35.2	0.1	-32.1	-31.5	0.2	-28.6	8.9	-6.0	8.1	-7.7	1.2	110	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	27.8	19.4	25.2	18.1	23.1	16.9	20.6	25.6	19.3	24.0	18.0	21.8	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.7	22.8	17.4	12.5	21.5	16.1	11.5	20.2	59.1	25.8	54.8	23.9	50.8	21.7	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.0	6.2	-35.0	30.5	3.1	1.8	-37.2	31.8	-39.1	32.8	-40.8	33.8	-43.1	35.1		
(5)			-34.9	22.4	3.1	1.7	-37.2	23.6	-39.1	24.6	-40.8	25.6	-43.1	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-12.0	-11.5	-6.1	0.6	7.2	12.8	16.3	13.4	8.6	2.6	-4.4	-8.4		
	DBStd	11.25	8.48	8.31	5.62	4.80	5.35	4.73	4.33	3.77	3.56	4.08	6.27	7.86		
	HDD10.0	3526	682	601	499	283	123	23	3	8	67	232	432	572		
	HDD18.3	6133	941	834	758	531	347	177	93	160	291	489	682	830		
	CDD10.0	482	0	0	0	2	35	106	198	113	27	1	0	0		
	CDD18.3	47	0	0	0	0	1	10	29	7	0	0	0	0		
	CDH23.3	443	0	0	0	1	21	111	246	63	1	0	0	0		
	CDH26.7	94	0	0	0	0	3	26	53	12	0	0	0	0		
(14) Wind	WSAvg	3.1	3.0	3.1	3.2	3.2	3.4	3.1	2.8	2.7	2.8	3.1	3.1	3.1		
(15) Precipitation	PrecAvg	578	36	29	29	29	44	59	66	72	57	65	49	43		
	PrecMax	735	66	56	50	72	81	101	136	145	120	127	75	74		
	PrecMin	460	8	8	9	1	14	13	4	19	16	23	24	20		
	PrecStd	82	13	13	10	16	18	23	38	31	25	24	13	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	3.1	6.9	19.1	26.1	29.8	30.8	28.2	20.8	13.2	7.0	3.8		
		MCWB	1.4	1.9	2.6	10.5	14.9	20.2	21.1	20.5	16.2	10.9	6.2	3.3		
	2%	DB	1.0	1.4	3.9	12.9	22.1	26.6	28.1	25.0	18.1	11.1	5.2	2.1		
		MCWB	0.6	0.8	1.4	7.8	13.5	17.8	19.7	18.7	14.0	9.6	4.7	1.6		
	5%	DB	0.0	0.4	2.6	9.4	19.2	23.9	26.1	22.2	16.0	9.4	3.9	1.3		
		MCWB	-0.3	-0.1	0.8	5.2	12.0	16.2	19.0	17.2	13.2	8.5	3.5	0.9		
	10%	DB	-1.2	-0.9	1.4	7.2	16.8	21.3	24.1	20.0	14.2	7.9	2.3	0.8		
		MCWB	-1.6	-1.6	0.2	3.8	11.2	15.0	18.2	16.2	12.1	7.2	1.8	0.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	2.1	3.1	11.2	17.3	20.9	22.8	21.6	16.9	11.7	6.4	3.1		
		MCDB	1.5	2.7	5.3	17.6	23.0	27.2	27.5	25.9	19.5	13.0	6.6	3.4		
	2%	WB	0.5	0.8	1.9	7.9	14.6	19.3	21.1	19.5	15.1	9.9	4.7	1.6		
		MCDB	0.7	1.2	3.1	12.1	19.5	25.3	26.3	23.6	17.4	10.8	5.1	1.9		
	5%	WB	-0.4	-0.1	1.1	5.7	13.0	17.4	19.9	17.8	13.7	8.5	3.3	0.9		
		MCDB	-0.1	0.3	2.1	8.4	18.3	21.5	24.6	21.0	15.6	9.3	3.6	1.3		
	10%	WB	-1.8	-1.7	0.3	4.1	11.3	15.6	18.7	16.5	12.4	7.1	2.0	0.2		
		MCDB	-1.5	-1.1	1.4	6.6	16.1	20.2	22.9	19.7	14.2	7.7	2.2	0.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.5	8.9	8.9	9.8	10.1	9.9	9.1	7.4	4.6	4.8	5.8		
		MCDBR	5.1	5.4	7.6	12.1	14.5	13.5	12.8	12.2	9.9	5.6	4.3	3.9		
	5% WB	MCWBR	5.1	5.1	5.8	7.6	7.7	6.6	6.5	6.6	6.5	4.7	4.2	3.9		
		MCWBR	5.1	5.5	5.9	11.0	13.0	11.8	11.5	10.8	8.6	5.4	4.0	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.227	0.287	0.316	0.352	0.357	0.350	0.388	0.364	0.337	0.311	0.356	0.222		
	taud		1.944	2.065	2.107	2.190	2.416	2.472	2.383	2.430	2.491	2.340	2.545	1.953		
	Ebn at Noon		293	589	747	812	852	867	815	802	751	584	283	92		
	Edh at Noon		27	64	96	112	99	96	102	89	68	50	23	10		
(44) All-Sky Solar Radiation	RadAvg		0.13	0.64	1.99	3.63	4.76	5.36	5.09	3.48	1.89	0.69	0.16	0.05		
	RadStd		0.01	0.12	0.26	0.34	0.44	0.50	0.52	0.40	0.20	0.11	0.03	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page